

Introduction to Computer Graphics

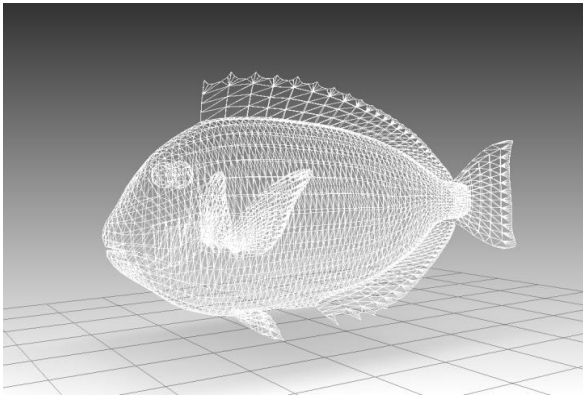
Computer Graphics
Instructor: Sungkil Lee

Big Picture of Computer Graphics (CG)

- **Computer graphics deals with:**

- all the aspects of creating images with a computer in hardware, software, and applications.

- **Three primary research areas in CG**



Modeling



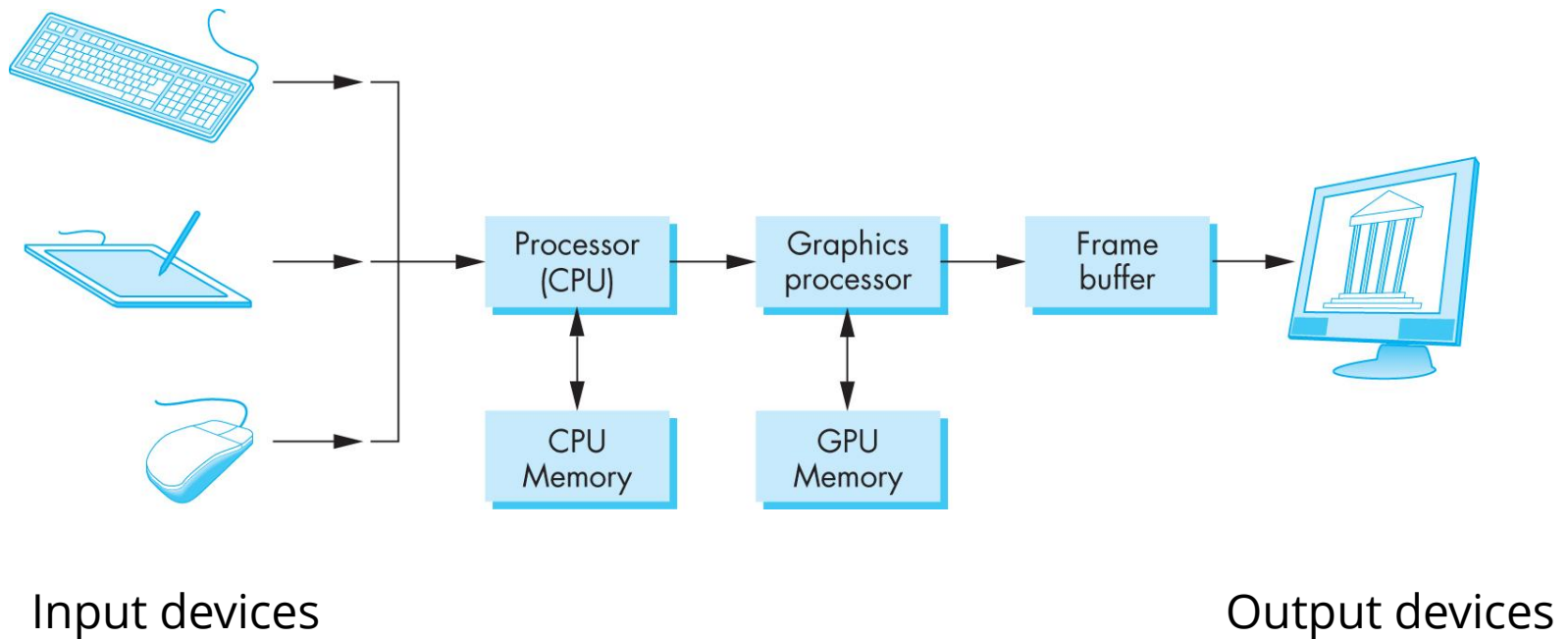
Rendering



Animation

Interactive Graphics System

- **Basic system for interactive graphics**
 - Recent mobile systems include touch-sensitive display for input.

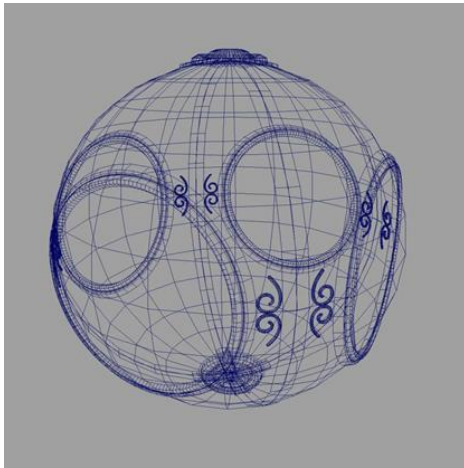


History

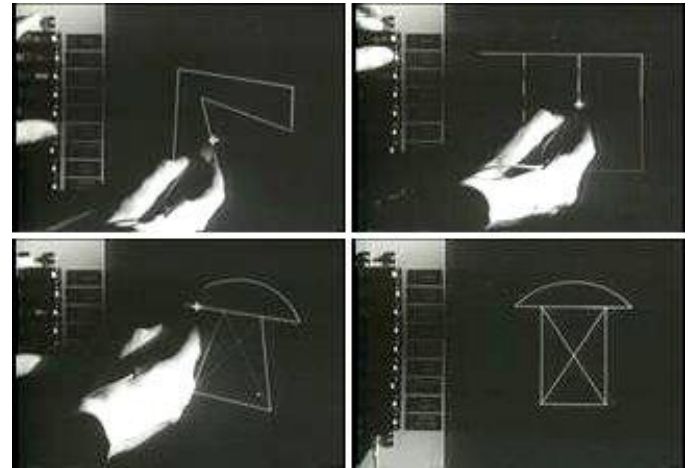
Brief History

- **1960s:**

- Raster graphics with wireframe display
 - Early predecessor of the modern raster graphics
- Sketchpad (Turing-awarded, 1988)
 - Software written by Ivan Sutherland (Ph.D. thesis at MIT).
 - The early concept of display loop, still common in computer graphics
 - A computer displays new images by light pen movements



Wireframe display

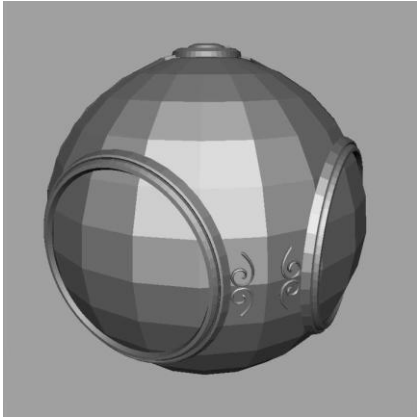


Sketchpad (Sutherland)

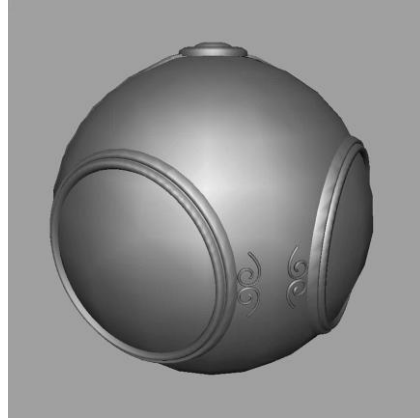
Brief History

- **1970-80s:**

- Raster graphics
 - Allows us to go from lines and wireframe images to filled polygons
 - Image produced as an array of picture elements (pixels) in the frame buffer
- **Hardware (VLSI) geometry engine by Silicon Graphics Inc. (SGI, 1982)**
 - Irix and **Iris GL**: Operating system and Graphics API of SGI



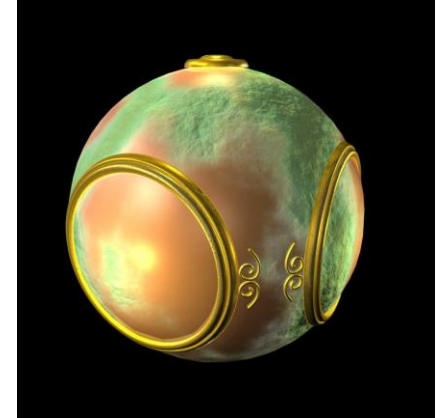
Early raster display



Smooth shading



Environment mapping



Bump mapping

- Industry-based standards
 - Pixar: RenderMan API (and REYES architecture)

Brief History

- **1990s:**

- Iris GL later became OpenGL (1992)
 - a platform-independent (no windowing support) rendering API
 - close enough to hardware to get excellent performance
- New hardware capabilities
 - Texture mapping, blending, accumulation, stencil buffers
- *Toy Story*:
 - the first completely computer-generated feature-length movie



Toy story, Pixar (1995)

Brief History

- **2000s-now:**

- Graphics cards for PCs dominate market
 - NVIDIA, ATI (now AMD), 3DLabs
- Game consoles and game players determine direction of market
 - Playstation, Wii, XBOX (+Kinect), ...
- Computer graphics software in movie industry
 - 3DS Max, Maya, Lightwave
- Display technology
 - LCD, PDP (Plasma display panel), LED, OLED, ...
- Programmable rendering pipelines
 - **GPU** named by NVIDIA

Applications:
What can we do with CG?



**Entertainment - Film Production (Toy Story, Pixar 1995)
First CG-Generated Full-Length Movie**



Entertainment - Games (Unreal Engine RTX Demo, Epic Games, 2018)
<https://www.youtube.com/watch?v=J3ue35ago3Y>



NVIDIA Marbles at Night RTX Demo (2020)
https://www.youtube.com/watch?v=NgcYLvlp_k



NVIDIA Racer X Real Time Simulation Tech Demo (2023)
<https://www.youtube.com/watch?v=GBYs4y1BtGg>

Metaverse

- **“Oasis” in Ready Player One (2018)**
 - <https://www.youtube.com/watch?v=cSp1dM2Vj48>





Training and Entertainment - Virtual Reality (Oculus VR)



Training and Simulation - Virtual Reality (Flight Simulation)

LLFF [Mildenhall 2019]



NeRF



Nearest Input

Neural Rendering: Neural Radiance Fields (NeRFs)
<https://www.youtube.com/watch?v=JuH79E8rdKc>



Example from

Differentiable Signed Distance Function Rendering
by Vicini et al. 2021 (SIGGRAPH'21)

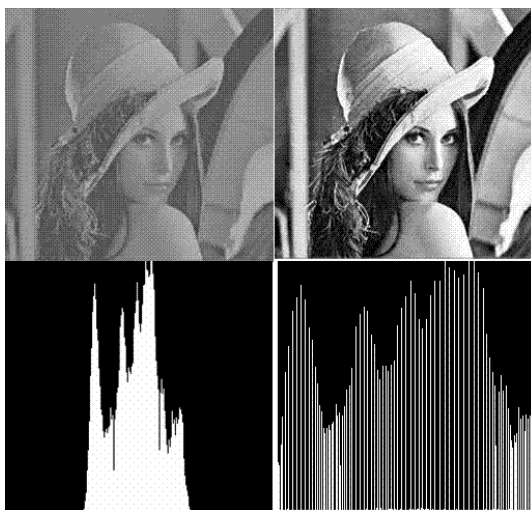


Neural Rendering: Differentiable Rendering
<https://www.youtube.com/watch?v=9Ja9buZx0Cs>

Related Areas

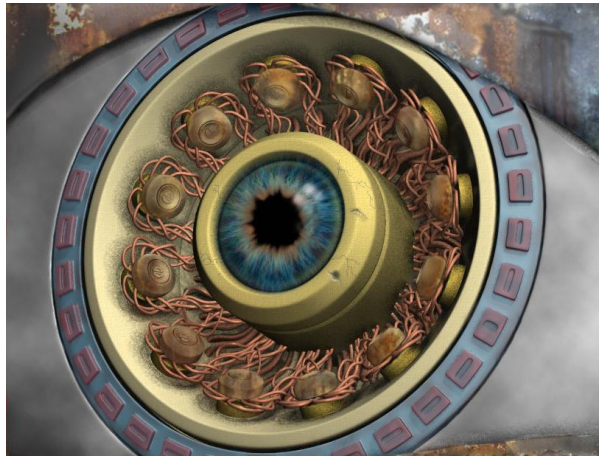
Image Processing (2D→2D)

- **Digital signal processing for which the input is a 2D/3D image, such as a photograph or video frame.**
 - The output of image processing may be either an image or a set of characteristics or parameters related to the image.



Computer Vision (2D→3D)

- **A field that includes methods for acquiring, processing, analyzing, and understanding images and, in general, high-dimensional data from the real world in order to produce numerical or symbolic information**
 - Also, called the machine vision or robot vision



Computer Graphics (3D→2D)

- **The study of digitally synthesizing and manipulating visual content. Although the term often refers to the study of 3D computer graphics, it also encompasses 2D graphics and image processing.**



Robotics

- **The study of the design, construction, operation, and application of robots, as well as computer systems for their control, sensory feedback, and information processing.**



Virtual Reality (VR)

- **Computer-simulated environments that can simulate physical presence in places in the real world, as well as in imaginary worlds.**
 - Simply, it's like simulation of five human sensory modalities.



Augmented Reality (AR)

- A live, direct or indirect, view of a physical, real-world environment whose elements are augmented by computer-generated sensory input such as sound, video, graphics or GPS data.



Metaverse

- **Metaverse (in Wikipedia)**

- is a collective *virtual shared space*, created by the convergence of virtually enhanced physical reality and *physically persistent* virtual space, including the sum of all virtual worlds, AR, and the Internet.
- The word “metaverse” is made up of the prefix “meta” (meaning beyond) and the stem “verse” derived from the universe.



Selected My Research Outcomes

Defocus Blur

- SIGGRAPH ASIA 2009: Depth-of-Field Rendering with Multiview Synthesis
- SIGGRAPH 2010: Real-time lens blur effects and focus control
- CGF: Real-Time Defocus Rendering with Level of Detail and Subsample Blur



Lens Flares

- *SIGGRAPH 2011: Physically-Based Real-Time Lens Flare Rendering*



Canon Zoom 70-200 mm f/11.0 (SIGGRAPH 2011)

3.3 frames/s

Lens Flares

- *EGSR 2013: Practical Lens Flare Rendering*

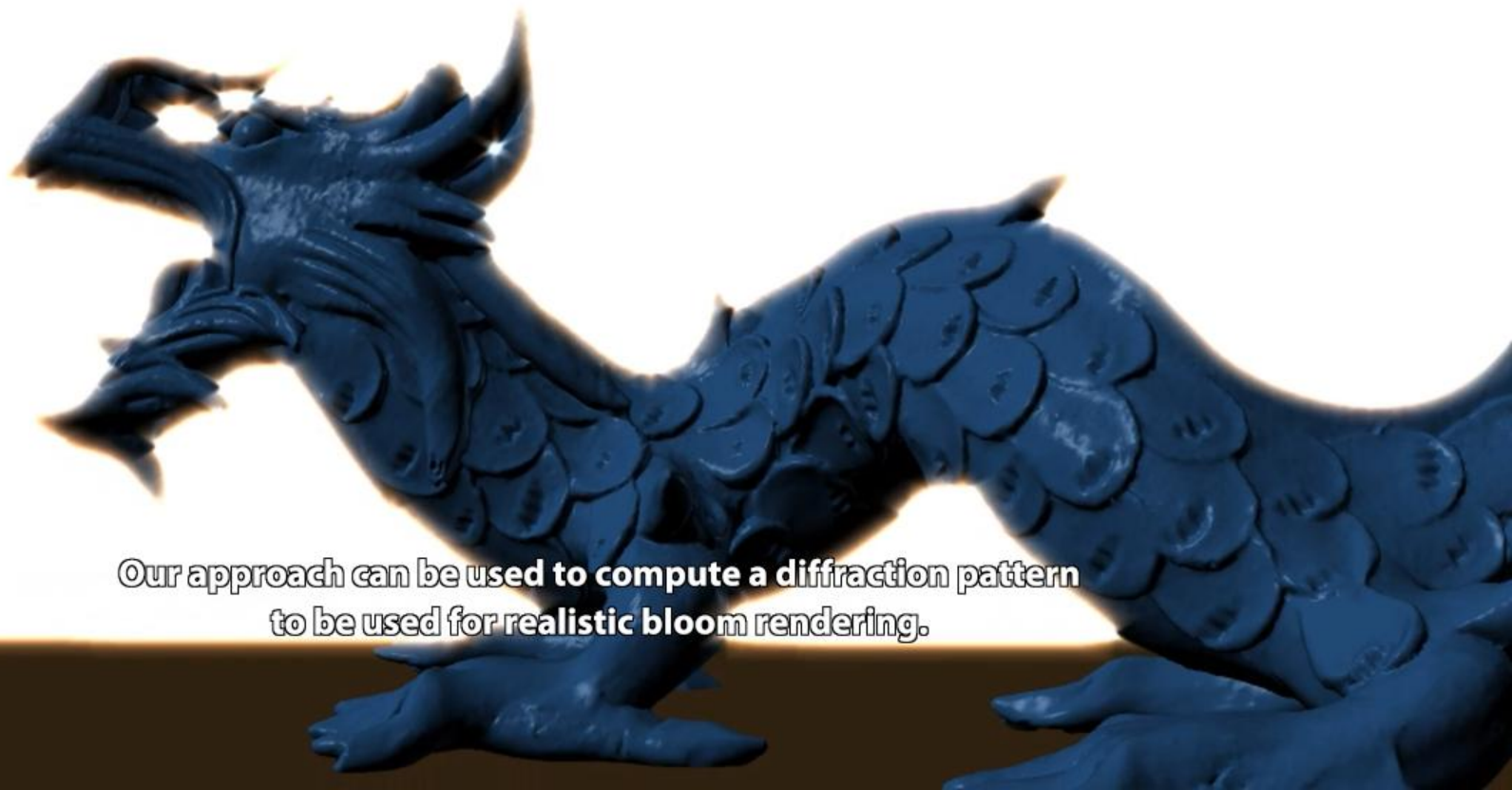


With our lens-flare effect
'Moonrise over Water', stockfootageforfree.com

Bloom

- *EGSR 2018: Quad-Based Fourier Transform for Efficient Diffraction Synthesis*

Realistic bloom rendering



Our approach can be used to compute a diffraction pattern
to be used for realistic bloom rendering.

Soft Shadows

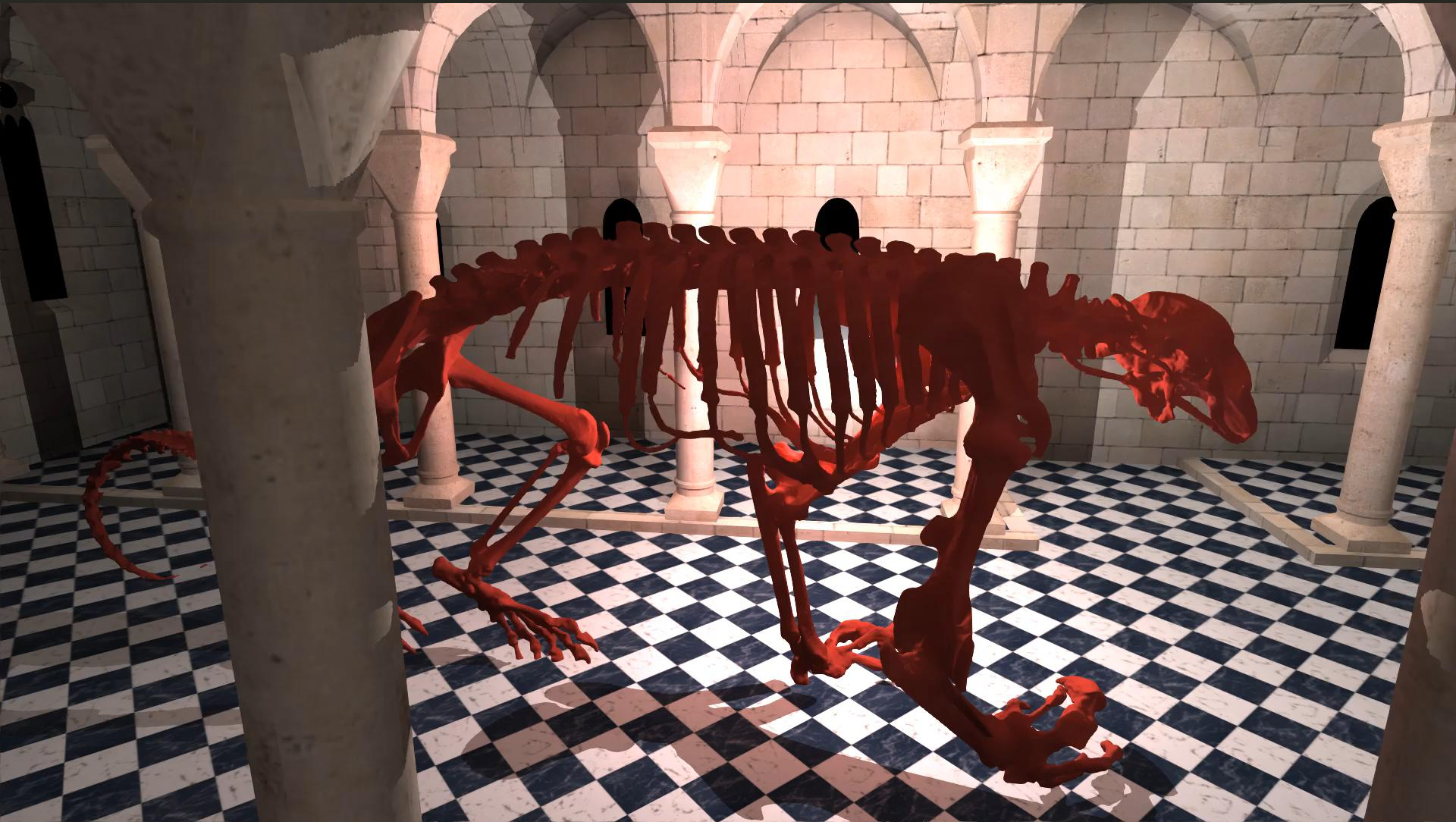
- TOG 2018 (presented at SIGGRAPH 2019): *Iterative Depth Warping*



MO scene with soft shadows

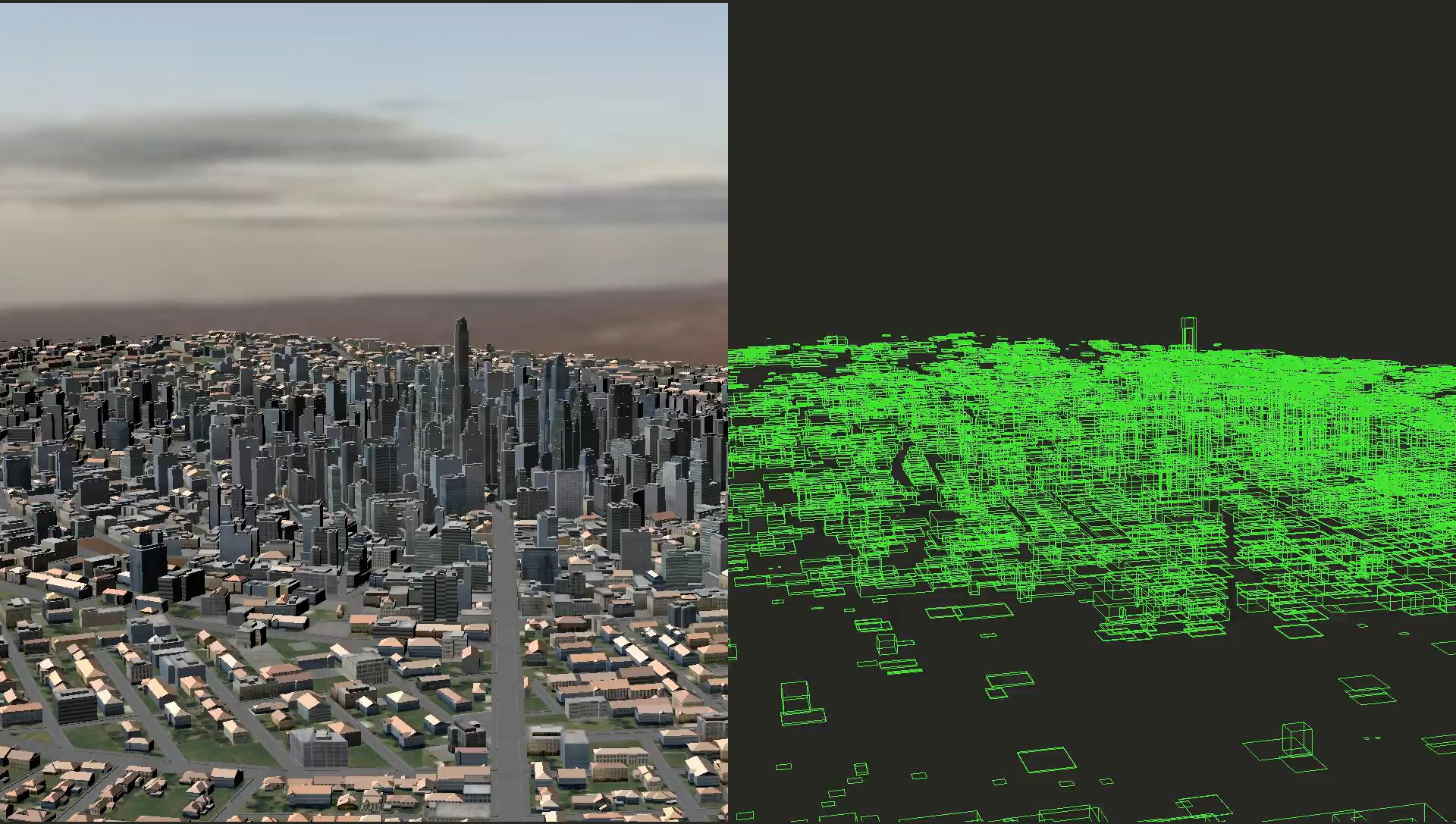
Real-Time Global Illumination

- CGF 2019: *MegaViews: Scalable Many-View Rendering with Concurrent Scene-View Hierarchy Traversal*



GPU Occlusion Culling

- *Eurographics 2021: Hierarchical Raster Occlusion Culling*



Dynamic Bokeh Rendering

- TVCG 2022: Real-Time Dynamic Bokeh Rendering with Efficient Look-Up Table Sampling



Novel-View Synthesis

- *ACM SIGGRAPH 2023: Potentially Visible Hidden-Volume Rendering for Multi-View Warping*

Potentially Visible Hidden Volumes (PVHVs)



Deep Learning for Depth/Defocus Estimation

- *CVPR 2019: Deep Defocus Map Estimation using Domain Adaptation*

